

ML Production Deployment and Strategies



Department Workflow



API and Microservice Management



Production Phase of the Workflow: Production Release



Deployment Strategies

Department Workflow

A generic workflow was introduced for Machine Learning Operations or MLOps. The workflow consisted of three modules: **Build**, **Deploy** and **Monitor**.

The deploy phase in MLOps workflow enables operationalization of the ML models created in the built step of the workflow. Once an ML model is trained, it is ready to be deployed in a production environment, exposing it to unseen data for making predictions. Production deployment integrates the model with the application to generate tangible business insights.

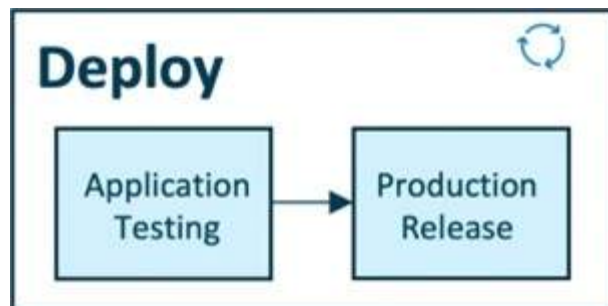
Some of the Model Deployment challenges:

- **Coordination Between Stakeholders:** Model deployment is a group task that needs constant communication among groups that may not have DS or ML backgrounds, such as DevOps, application developers, and business team members.
- **Programming Language Discrepancy:** ML models are build using (Python/TensorFlow/PyTorch) compared to languages developers use (Java/C++/Ruby)
- **Model Drift:**Model drift happens when the model's prediction performance degrades below the accepted benchmark.
 - Changing behavior of new data
 - Changing interpretation of the new data
- **On-Prem vs. Cloud-Based Deployment:** Using on-premise resources or opt for cloud services. The DS and infrastructure teams should consider all the pros and

cons before deciding on a production strategy.

- **Clear Ownership:** Clear definition of responsibilities and coordination of activities between the data scientist's team and the DevOps team.
- **Model Performance Monitoring:** A monitoring framework needs to be in place to track the model's performance consistently.
 - When to retrain the model again?
 - How much data to use to retrain the model?
 - What is the performance benchmark for the model?
- **Release/Version Management:** Version control for models, their different file dependencies, and data resource pointers.
- **Privacy-Preserving and Secure Model:** safeguard the models from any adversarial attacks
- **Scalability:** A proper infrastructure, tools, and services to efficiently respond to increased user requests.

Deployment Phase of the Workflow: Application Testing



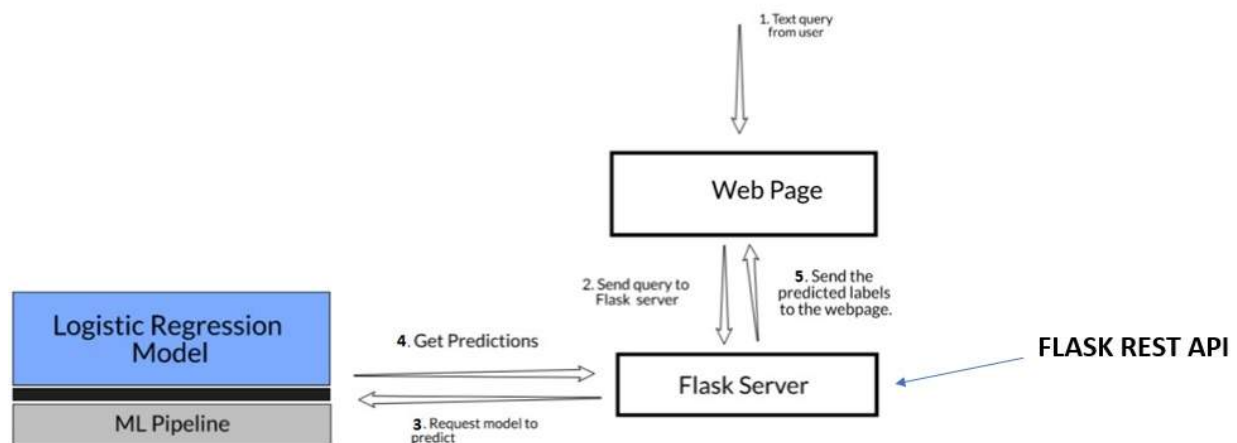
The first stage in production deployment is to deploy the trained ML model in the production testing phase where all the trained models are tested for robustness, performance, and behavior in a production-like (test) environment called application testing.

In the application testing phase, the test environment (pre-production), replicates the production environment.

The ML model for testing is deployed as an API or streaming service in the test environment to deployment targets such as the web localhost machine, container instances, Kubernetes cluster, or scalable virtual machines or edge devices as per the need and use case.

A standard testing and production deployment approach is to expose the ML model as REST API endpoints to serve the requests within the application platform. We will do this in the next lesson

- ML applications can be deployed as APIs
- API is short for Application Programming Interface. It's simply an intermediary between two independent subsystems or applications that communicate with each other.



API and Microservice Management

An API is a gateway that enables developers to communicate with an application. APIs enable two things:

- Access to an application's data
- The use of an application's functionality

The Representational State Transfer (REST) protocol has become the standard protocol to use for many applications

API Protocol	Release Year	Formatting Type	Key Strength
SOAP	1990's	XML	Widely used and established
REST	2000	XML, JSON, and others	Flexible data formatting
JSON-RPC	2005	JSON	Easy implementation
Thrift	2007	JSON or binary	Adaptability to many use cases
gRPC	2015	JSON and others	Ability to define any type of function
GraphQL	2015	JSON	Flexible data structuring

A REST API or RESTful API is based on REST. Most companies prefer developing their ML applications based on the REST API protocol. A RESTful API uses existing HTTP

methodologies for communication. The fundamental HTTP methods are GET, POST, PUT, PATCH, and DELETE. These methods correspond to CRUD operations such as create, read, update, and delete. Using these methods, RESTful APIs are developed to serve ML models. There are many different technologies for REST API development, such as Flask, Django, Streamlit, FastAPI, etc.

In the next lesson, we will use FLASK to deploy a machine learning model on a local web server (localhost). Before fully deploying the ML application to the production server, the application must be tested in the test environment (as a REST API) first. The ML application performance is monitored on data sampled from a production environment, not the model training data. Model inference in batch or streaming tests the model deployed in the test environment for robustness and performance.

When the ML model's performance meets the test environment standards, it can be deployed to the production environment in real time for business decision-making.

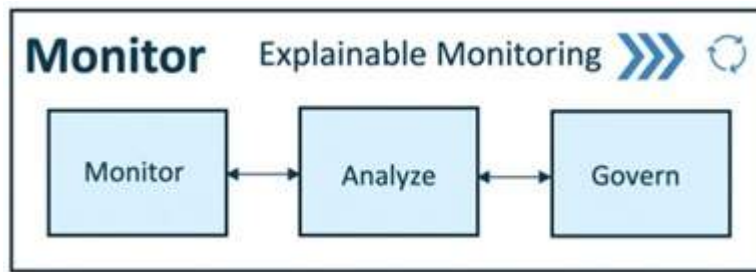
Production Phase of the Workflow: Production Release



Production release: Previously tested and approved models are deployed in the production environment for model inference to generate business or operational value.

This product release is deployed to the production environment using CI/CD (continuous Integration /Continuous Delivery) pipelines. We will cover CI/CD using GitHub Actions later in this module.

Monitoring Phase of Workflow



The last stage in the MLOps Workflow is explainable monitoring. This module is made of three stages:

1. The Monitor stage captures information to monitor data integrity, model drift, and application performance. Use metrics such as accuracy, F1 score, precision, recall tracking data integrity, and model drift.
2. The Analyze stage ensures optimal operations regarding the performance of models in production systems in correlation to business decisions or impacts. Evaluate essential aspects of the model performance, such as model fairness, trust, bias, transparency, and error analysis to improve the model.
3. The Govern stage is where the system generates certain alerts and actions to govern the system.

Quality assurance expert gets alerted when model performance deteriorates (for example, low accuracy, high bias, and so on) below a pre-defined threshold. The product owner initiates a trigger to retrain and deploy an alternative model.

An essential aspect of governance is "compliance" with local and global laws and rules. For compliance, model explainability and transparency are vital. Model auditing and reporting provide end-to-end traceability and explainability for production models.

Let's go back and further examine the explainable monitoring module and the model retaining/redeployment actions that might be taken. Once the model starts performing below the expected benchmark level, the model needs to be retrained and re-evaluated to replace the old model with the new.

This includes model management (versioning, features, etc.) Machine learning models need to be deployed frequently and continuously to ensure expected

performance; hence we need a deployment strategy.

Ideally, the model is deployed again in production without affecting the existing user requests.

Deployment Strategies

A deployment strategy is a way to update (retrain) an existing ML Model in production. Before deciding on a deployment strategy, a few questions must be considered:

- What is an acceptable downtime (SLA: Service Level Agreement) for the ML solution?
- How to roll back the model in case the new model deployment has issues?
- Any requirements on backward compatibility of the ML solutions?
- Approximate operation cost and overheads?

The different model deployment strategies are:

Recreate deployment: In this strategy, the operation is taken down, then brought back up with the new model—a good plan for when the downtime of the ML solution is not a concern.

- Advantages:
 1. One version of the model needs to be managed at any time given time.
 2. Deployment risk is reduced.
 3. Backward compatibility is not needed

- Disadvantages:

1. Downtime for the production system
2. Slow deployment

Canary deployment: Rollouts the new version of the model incrementally. The latest version exists alongside the old one; the most recent version is tested before scaling it out completely. A percentage of prediction requests are forwarded to the new model. Suitable for when models need to be evaluated against production data before scaling out the deployment with zero downtime

- Advantages:

1. Ability to evaluate the new model against the production data.
2. No downtime.
3. Fast rollbacks by redirecting the users back to the older model version, 3) Least risk-prone deployment strategy.

- Disadvantages:

1. Testing the new version happens in production.
2. Slow rollouts since testing in production can take time.
3. Backward compatibility is required as multiple versions of the models are running in production at the same time.

Blue-Green deployment: Blue-Green deployment is a progressive deployment strategy with two production environments. The **Blue** environment has the current model version deployed and is live to serve the current request. The **Green**

environment acts as a staging environment that contains the new model version to perform tests to ensure expected performance and functionality.

Once the testing is complete on the Green environment, the traffic will be diverted from the Blue environment to the Green environment making it live.

Use when the application cannot afford any downtime. It can maintain the backward compatibility of the models, and the operational cost overhead is justified. The blue environment can be maintained for a possible rollback, decommissioned, or used for the next deployment.

- Advantages:

1. No downtime.
2. Rollbacks are straightforward and reliable, and quick.
3. Reduced deployment risk

- Disadvantages:

1. Increased operational cost due to maintaining two environments.
2. Backward compatibility is required.

A/B testing deployment: In A/B testing deployment, different model versions are deployed to production (could be more than two versions). Strategy compares the performance results among them and selects the model that performs the best. Divide the users into groups based on the number of models being compared. Use when two models are doing reasonably well and must decide which to keep.

- Advantages:

1. Quick results, Low-performing models can be eliminated by monitoring statistically significant performance and not waiting until the experiment's end.
2. Allows models to be rolled out to production easily

- Disadvantages:

1. Complex setup can predict unreliable results if a low-performing model is used for A/B testing.